

MATRIX ACADEMY

- 5/11, Rajender Nagar, Sec-5, Opp. DAV Public School, Sahibabad, Ghaziabad, U.P.
- S-14, First Floor, Opp. S.M. World, Shalimar Garden-II, Sahibabad, Ghaziabad, U.P.

Name Shivansh Sharma Class & Batch Xth 3-4pm Subject Maths Date 23/08/26

Section-A

Q1.

$$D = b^2 - 4ac$$

for real & distinct roots, $D > 0$

$$\Rightarrow b^2 - 4ac > 0$$

$$\Rightarrow (-8)^2 - 4(1)(k) > 0$$

$$64 - 4k > 0$$

$$+4k < +64$$

$$k < 16$$

Ans $\rightarrow$ D) $k > 16$ X

3

$$\begin{array}{r} 2 \overline{) 32} \\ \underline{2} \\ 16 \\ \underline{16} \\ 0 \end{array}$$

$$\begin{array}{r} 750^\circ \end{array}$$

Q2.

$$D = b^2 - 4ac$$

for rational roots, $D > 0$

$$b^2 - 4ac > 0$$

$$(k)^2 - 4(2)(-4) > 0$$

$$k^2 + 32 > 0$$

$$k^2 > -32$$

$$k > \pm \sqrt{32}$$

(C) $+2$ X

Q3. (c) $BC \parallel XY$ ✓

Q4. (b) 27 feet ✓

Q5. (A) Both A & R are true and R is correct explanation of A. ✓

Section-B



Let the breadth of rectangular field is x
 then it's length $\rightarrow 3x$
 $\text{Area} = 147 \text{ m}^2$

Ans: $(m)(3m) = 147$

$3x^2 - 147 = 0$

$3(x^2 - 49) = 0$

$x^2 - 49 = 0$

$(x-7)(x+7) = 0$

$x = 7 \text{ m}$ or $x = -7 \text{ m}$

$\Rightarrow$ $x = -7 \text{ m}$ is rejected, as breadth can't be negative.

$\frac{2.6}{3.9} = \frac{3}{5.1}$

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$\frac{18}{2} = \frac{3}{2}$

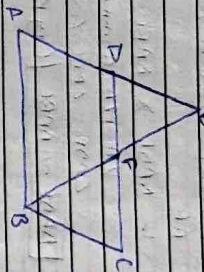
$x = \frac{3 \times 18}{2}$

$x = 27$

Q7

$\Rightarrow$ breadth $\Rightarrow 7 \text{ m}$

$\therefore$ length $\Rightarrow 3(7) = 21 \text{ m}$



proof:

Since $ABCD$ is a line, $\angle ADE = \angle B$ (alt. angles)

$\therefore$ AE and BE are produced to F

$\therefore$ AE and BE are also not to CB

now in $\triangle ABE$ & $\triangle CFE$

$\angle AEB = \angle CEF$ (alt. angles)

$\angle EAB = \angle ECF$ ($\therefore$ opp. angles are equal)

$\therefore \triangle ABE \sim \triangle CFE$ (by AA)

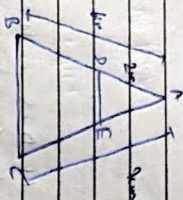
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Date 23/06/20

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Q8.



Given: In $\triangle ABC$, $DE \parallel BC$.

$AD = 3 \text{ cm}$, $BC = 8 \text{ cm}$, $AC = 9 \text{ cm}$

to find: AE

Solution: In $\triangle ADE \sim \triangle ABC$ -

(A-A (common))

$\angle ADE = \angle ABC$ (Corresponding $\angle$ s)

$\Rightarrow \triangle ADE \sim \triangle ABC$ (by AA)

$\Rightarrow \frac{AD}{AB} = \frac{AE}{AC}$ (CPST)

$\frac{3}{9} = \frac{AE}{9}$

$\Rightarrow \frac{2}{9} = \frac{AE}{9}$

$\frac{1}{3} = \frac{AE}{9}$

$\frac{1}{3} = \frac{AE}{9}$

$AE = 3 \text{ cm}$

Q9.

Q/R of object $\rightarrow 75$

dist. $\rightarrow 300 \text{ km}$

Let original speed $= x \text{ km/hr}$

$\frac{300}{x} = \frac{300}{x+5}$

$75 - x = x^2$

$x^2 + 5x - 75 = 0$

$x^2 + 15x - 75 = 0$

$x^2 + 20x - 75 = 0$

$x^2 + 25x - 75 = 0$

$x^2 + 30x - 75 = 0$

$x^2 + 35x - 75 = 0$

$x^2 + 40x - 75 = 0$

$x^2 + 45x - 75 = 0$

$x^2 + 50x - 75 = 0$

$x^2 + 55x - 75 = 0$

$x^2 + 60x - 75 = 0$

$x^2 + 65x - 75 = 0$

$x^2 + 70x - 75 = 0$

Section-1)

5/7500

5/1300

5/250

5/50

5/10

5/2

5/1

5/150

5/150

5/150

5/150

5/150

5/150

5/150

5/150

5/150

5/150

5/150

$$\Rightarrow n(n+30) - 25(n+30) = 0$$

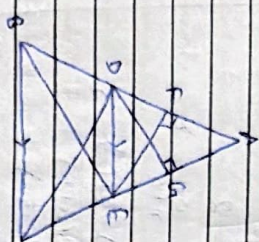
$$\Rightarrow (n+30)(n-25) = 0$$

$$\Rightarrow \boxed{n = -30} \text{ or } \boxed{n = 25}$$

As rejected as
Speed cannot be
negative

$\therefore$ the usual speed of train was 25 km/hr Ans.

Q11.



to prove - $\frac{AD}{BD} = \frac{AE}{EC}$
given - In $\triangle ABC$, $DE \parallel BC$

construction - Join BF & CD

draw $EF \perp AD$ & $DG \perp AE$

$$\text{proof - } \frac{\text{area of } \triangle ADE}{\text{area of } \triangle BDE} = \frac{\frac{1}{2} \times AD \times EF}{\frac{1}{2} \times BD \times EF}$$

$$\Rightarrow \frac{\text{area } \triangle ADE}{\text{area } \triangle BDE} = \frac{AD}{BD} \quad \text{--- (1)}$$

$$\text{now, } \frac{\text{area } \triangle ADE}{\text{area } \triangle CEB} = \frac{\frac{1}{2} \times AE \times EF}{\frac{1}{2} \times EC \times EF} = \frac{AE}{EC} \quad \text{--- (2)}$$

now, we know that two triangle with same base & equal height have the same area.

$$\Rightarrow \text{area } \triangle CEB = \text{area } \triangle BDE$$

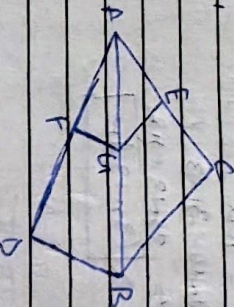
$$\Rightarrow \therefore \frac{\text{area } \triangle ADE}{\text{area } \triangle BDE} = \frac{\text{area } \triangle ADE}{\text{area } \triangle CEB}$$

from (1) & (2)

$$\boxed{\frac{AD}{BD} = \frac{AE}{EC}}$$

hence proved.

now,



given - $EF \parallel AD$ & $EF \parallel DB$
to prove - $\frac{AE}{EC} = \frac{AF}{FB}$

proof - In $\triangle ABC$, $EF \parallel DB$, so by BPT -

$$\frac{AE}{EC} = \frac{AF}{FB} \quad \text{--- (1)}$$

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Date: 23/09/21

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In $\triangle ABC$, $\triangle ABD$, So by BPT -

$$\Rightarrow \frac{AF}{FD} = \frac{AG}{GD} \quad (i)$$

from $\triangle ABC$

$$\Rightarrow \frac{AE}{EC} = \frac{AF}{FD}$$

hence proved.

Section-E

Q11 (i) length of square (glass board) = n m

then the length of glass = $3n+3$
~~then~~ $\Rightarrow$ total length of glass = $3n+3+n$

$$\Rightarrow (3n+3) \text{ m}$$



(ii)

perimeter of field $\Rightarrow 2(l+b)$

$$\Rightarrow 2[3n+3+n]$$

$$\Rightarrow 2[4n+3]$$

$$\Rightarrow [8n+6] \text{ m}$$

(iii)

area of total glass = 1260 m^2

$$\Rightarrow l \times b = 1260 \text{ m}^2$$

$$\Rightarrow (3n+3)(n) = 1260$$

$$\Rightarrow 3n^2 + 3n = 1260$$

$$\Rightarrow 3n^2 + 3n - 1260 = 0$$

$$\Rightarrow 3(n^2 + n - 420) = 0$$

$$\Rightarrow n^2 + n - 420 = 0$$

$$\Rightarrow n^2 + 21n - 20n - 420 = 0$$

$$\Rightarrow n(n+21) - 20(n+21) = 0$$

$$\Rightarrow (n+21)(n-20) = 0$$

$$\Rightarrow [n = -21] \text{ or } [n = 20 \text{ m}]$$

rejected

$$\therefore [n = 20 \text{ m}]$$

2	200
2	105
5	35
5	7
1	1

Section C

Q9.

$$SR = ₹75$$

$$Profit = 75 - x$$

$$0.15x, Profit = x \cdot (1 - 0.15) = x \left(\frac{15}{100} \right)$$

5

$$\Rightarrow \frac{x(15)}{100} - 75 - x$$

$$\Rightarrow \frac{x^2}{100} + x - 75 = 0$$

$$\Rightarrow x^2 + 100x - 7500 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Rightarrow x = \frac{-100 \pm \sqrt{(100)^2 - 4(1)(-7500)}}{2(1)}$$

2(1)

$$\Rightarrow x = \frac{-100 \pm \sqrt{10000 + 30000}}{2}$$

2

$$x = \frac{-100 \pm \sqrt{40000}}{2}$$

$$x = \frac{-100 \pm 200}{2}$$

$$x = \frac{-100 - 200}{2}$$

2

$$x = \frac{-100 + 200}{2}$$

$$x = \frac{-300}{2}$$

$$x = \frac{100}{2}$$

$$\boxed{x = -150}$$

$$\boxed{x = 50}$$

rejected

$\therefore$ The cost price of the article is ₹50.